

MADHUMANTI SAHA

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SUMMARY

B.Tech graduate with strong foundations in Java, Python, SQL, and SDLC/STLC principles. Experienced in designing robust application architectures, integration workflows, and handling extensive data validation using REST APIs and Postman. Passionate about software quality, debugging, and leveraging programming fundamentals to build and maintain efficient automated testing frameworks.

EDUCATION

Academy of Technology

Bachelor of Technology (Electrical Engineering), CGPA: 7.29/10 (Up to 7th Sem)

2022 – 2026

Hooghly, WB

PROJECTS

JobMatch - Online Job Portal | [GitHub](#) | Java, Spring Boot, Spring Security, JWT, MySQL, JPA, React

- Built an automated platform that correlates applicant data with open roles to optimize the recruitment matching pipeline.
- Engineered a matching algorithm/search functionality to process candidate qualifications and automatically filter or rank suitable career positions, improving the efficiency of the job search experience.
- Executed functional, integration, and system testing on user authentication modules and data pipelines; utilized Postman to perform rigorous API testing on all REST endpoints to ensure data integrity and track software defects.
- Designed a responsive and intuitive user interface along with secure backend data stores to seamlessly manage job descriptions, user resumes, and application workflows.

BriefURL - URL Shortener & Analytics Platform | [GitHub](#) | Java, Spring Boot, Spring Security, JWT, MySQL, JPA

- Developed a high-performance URL shortening service using Spring Boot 3 and Java 23, architecting a relational database schema in PostgreSQL to manage complex one-to-many relationships between users, mapped links, and click events. Built a custom 8-character slug generation algorithm to ensure unique, high-collision-resistant short URLs for efficient sharing
- Engineered a secure authentication layer using Spring Security and JWT, implementing custom filters to intercept and validate request headers. Bridged the gap between the application's custom user entity and Spring Security's interface to support production-grade authorization and session-independent private routing
- Implemented a real-time tracking system that captures granular click events and increments global counters during 302 redirects. Leveraged the Java Stream API and JPA between-clause queries to aggregate and transform raw database records into date-wise analytics, providing users with detailed engagement insights.
- Containerized the backend service using Docker and orchestrated a multistage build process to ensure environment consistency. Achieved production readiness by externalizing critical configurations via environment variables (.env) and deploying the containerized API to Render, integrated with a Neon Cloud database for persistent storage.

TECHNICAL SKILLS

Languages: Java, Python, SQL, HTML, CSS

Backend: Spring Boot, Spring Security, REST APIs, JWT, JPA, Hibernate, Postman

Databases: MySQL, SQL Server, MongoDB

Core CS: Data Structures Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Testing Tools: SDLC, Manual Testing Concepts, API Testing (Postman), Bug Tracking, Git, GitHub

CERTIFICATIONS

Salesforce Developer Virtual Internship: SmartBridge (AICTE & Salesforce) (2025)

Worked on workflow automation, data management, and business process optimization using Salesforce tools. [\[Verify\]](#)

Oracle Certified Associate: MySQL Implementation Certified Associate. [\[Verify\]](#)

ACHIEVEMENTS

Smart India Hackathon (SIH) 2024: Recognized among the Top 8 teams in the national competition.

RC Car Competition Winner: Led a 5-member cross-functional team to secure 1st place in a college-level racing competition through effective coordination and decision-making.

IEEE Student Member: Active member of IEEE Student Branch (AOT), participating in technical seminars, workshops, and collaborative engineering discussions.